Fiduccia–Mattheyses refinement

FM refines a bipartition with single-vertex moves, locking

Same bad seed, different move set

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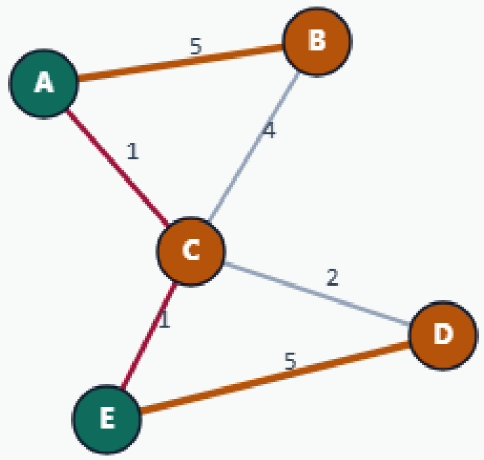
Same bad seed, different move set

Step 1 / 5 · bad-seed · module02-07-fiduccia-mattheyses



Explanation

FM starts from the same cutsize-12 seed, but moves one vertex at a time instead of swapping a pair. That suits hypergraph / cell-move implementations.

- Single-vertex moves across the cut
- Bucketed gains for speed (classic FM)
- Balance tolerance limits lopsided moves

```
seed cutsize: 12
parts: AE|BCD
```

Same bad seed, different move set

Move 1: flip D (gain 3)

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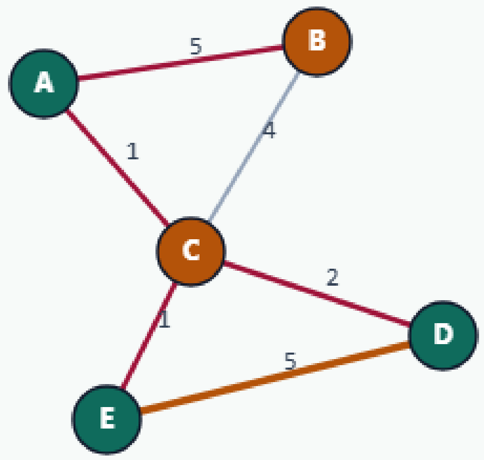
Move 1: flip D (gain 3)

Step 2 / 5 · move-d · module02-07-fiduccia-mattheyses



Explanation

Highest legal move sends D to the other side with gain 3. Partial progress: D joins E’s side early.

- Pick unlocked vertex with best gain
- Apply move, lock vertex
- Update neighbor gains

```
move: D(g=3)
running toward best prefix
```

Move 1: flip D (gain 3)

Move 2: flip A (gain 6)

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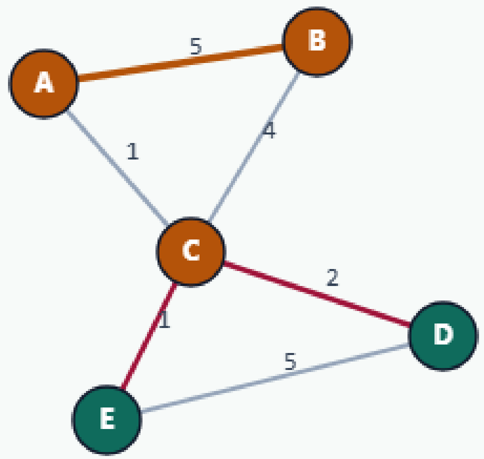
Move 2: flip A (gain 6)

Step 3 / 5 · move-a · module02-07-fiduccia-mattheyses



Explanation

Next, A flips with gain 6. Cumulative gain is 3+6=9—the same total improvement KL found with one swap.

- Two moves ≈ one KL swap's worth of gain
- best_k=2 keeps both moves
- bestCum=9

```
moves: D(3), A(6)
bestCum=9
cut 12→3
```

Move 2: flip A (gain 6)

Final ABC|DE, cutsize 3

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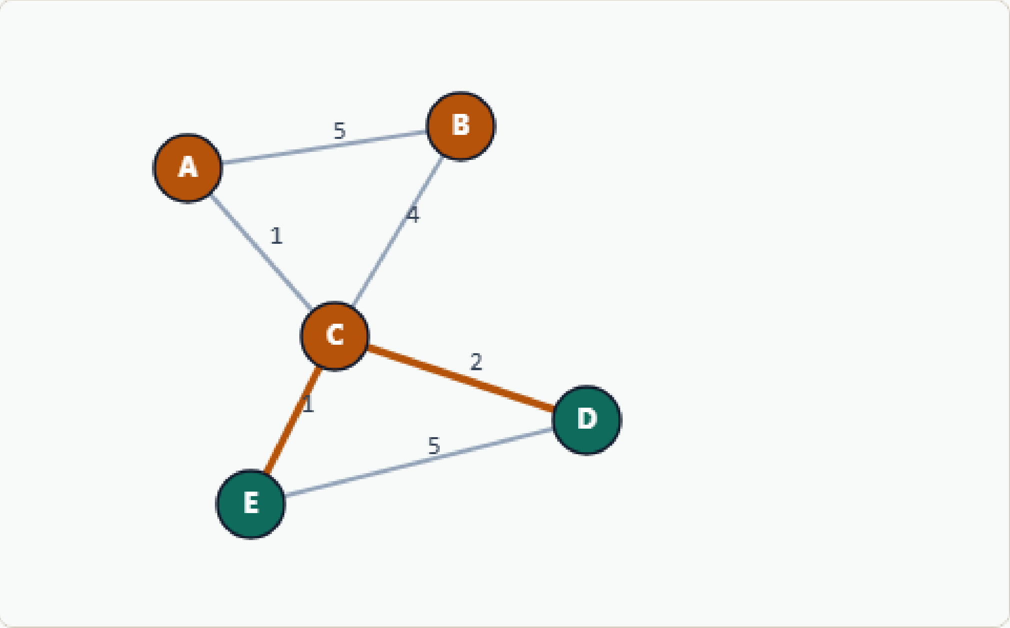
Final ABC|DE, cutsize 3

Step 4 / 5 · final · module02-07-fiduccia-mattheyses



Explanation

FM lands on the same refined bipartition as KL. Teaching point: move style differs, destination quality matches on this instance.

- parts: ABC|DE
- cutsize: 3
- Compare with KL transcript side-by-side

```
cutsize: 3
pass 0 improved=true
```

Final ABC|DE, cutsize 3

Pass 1 confirms local optimum

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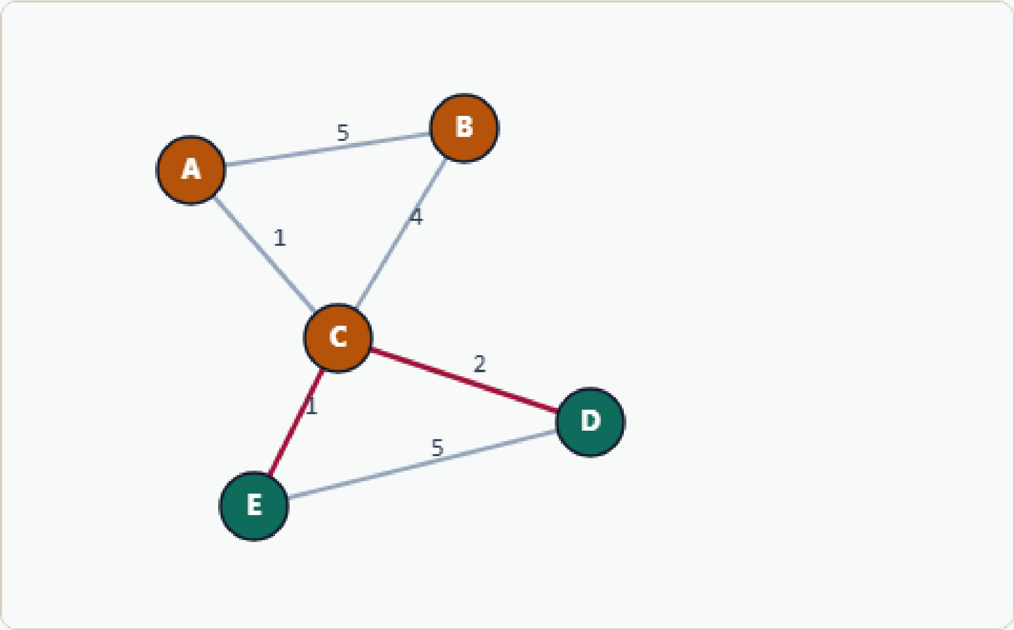
Pass 1 confirms local optimum

Step 5 / 5 · pass1-stop · module02-07-fiduccia-mattheyses



Explanation

A second FM pass finds no improving move prefix. Stop. In real tools, FM often runs inside multilevel V-cycles with tighter balance and hyperedges.

- improved=false on pass 1
- Cell moves scale to large netlists
- Next courses: multilevel + hypergraph

```
pass 1: best_k=0
stop
```

Pass 1 confirms local optimum

Browser lab track

- In the browser lab, show the seed, run FM, and compare the move list with KL's single swap
- Clear the challenges for twelve-to-three and the D-then-A prefix

Implement track

- Run FM on the shared bad seed
- Confirm moves D then A in the accepted prefix, and cut twelve to three
- Re-implement selection and rollback until tests pass

Implement track — try these

```
# FM refinement on the shared bad seed
export PYTHONPATH=../common
python ../common/solvers.py examples/tiny_graph.json --mode fm
--seed ../module02-05-kernighan-lin/examples/seed_partition.json
```

Pitfalls to watch

- Skipping rollback keeps a worse locked state
- Ignoring balance lets one side empty
- Stale gains after a move pick the wrong cell next

Your turn

- Reproduce twelve to three